

RiskPulse AI

Financial Risk Intelligence Engine
+ Strategic Portfolio Stress Testing

(S&P Global & CRISIL Campus Hackathon 2026)

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•THE PROBLEM

Financial risk often appears first in unstructured information:

- Breaking news
- Market commentary
- Social-media signals

The challenge is converting this information into actionable, structured risk signals quickly.

OUR APPROACH

RiskPulse AI:

1. Ingests public/synthetic text data
2. Performs sentiment analysis
3. Classifies financial events
4. Calculates impact and risk scores
5. Triggers portfolio stress testing
6. Visualizes portfolio impact

DATA SOURCES

GDELT News

NewsAPI (Optional)

Synthetic Social Feed

Synthetic Portfolio



DATA NORMALIZATION



NLP RISK ENGINE

- VADER Sentiment
- Event Classification
- Impact Scoring
- Risk Scoring



RISK SIGNALS



STRESS-TEST TRIGGER



SCENARIO SHOCK ENGINE



PORTFOLIO IMPACT



STREAMLIT DASHBOARD

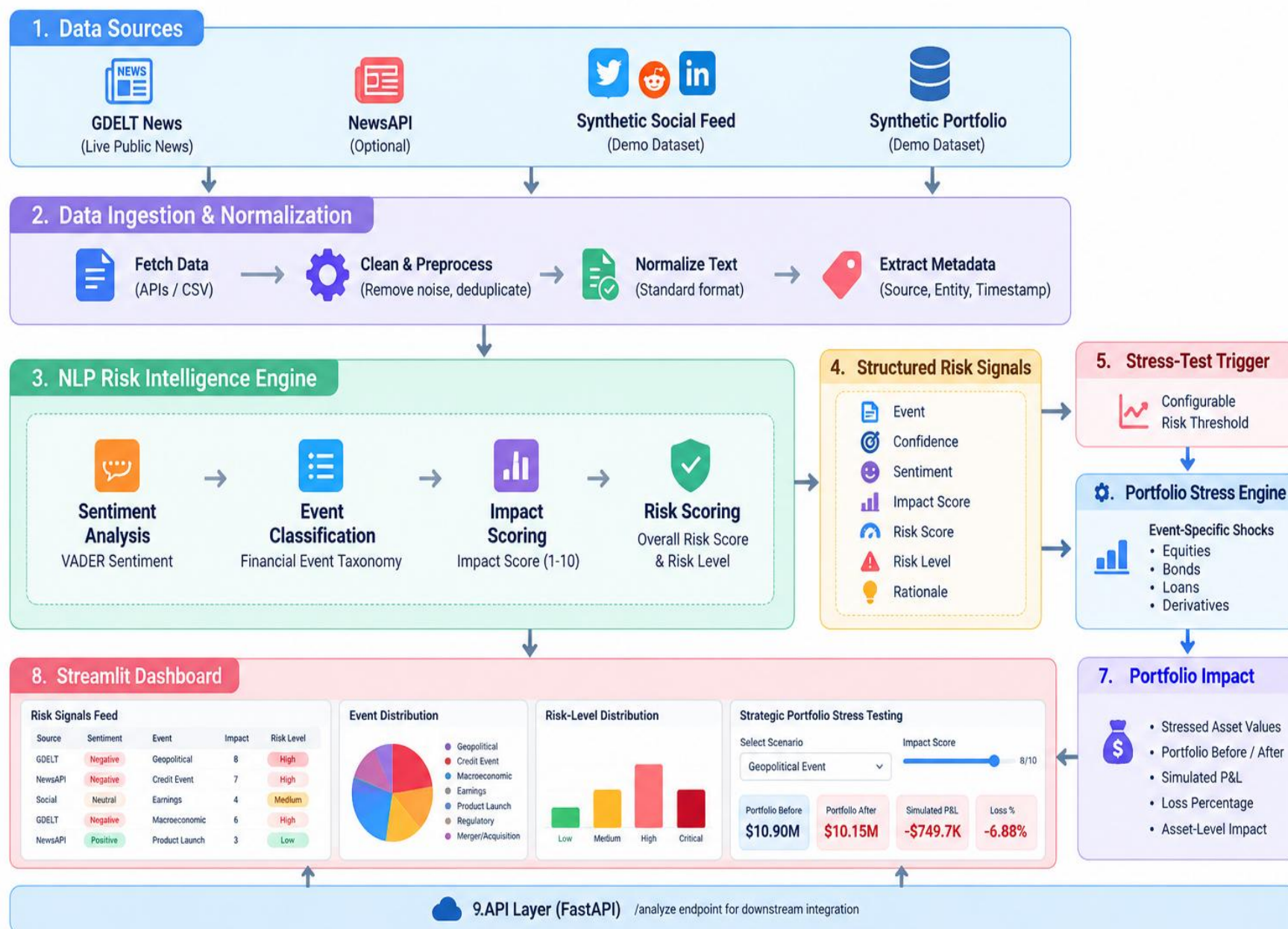


FASTAPI ENDPOINT



RiskPulse AI | System Architecture

Financial Risk Intelligence Engine + Strategic Portfolio Stress Testing



Key Results

Screenshot 1: the Strategic Portfolio Stress Testing screen

2. Module B — Strategic Portfolio Stress Testing

Scenario Event

Geopolitical

Scenario Impact Score

8

 Stress Trigger ACTIVE — Impact Score = 8

 Run Stress Test

Portfolio Before

\$10,900,000

Portfolio After

\$10,150,286

Simulated P&L

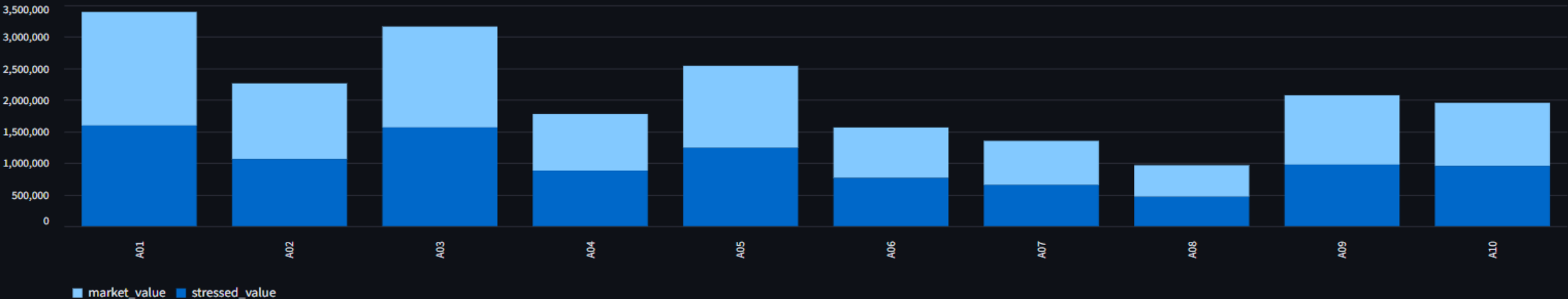
\$-749,714

Loss %

-6.88%



Portfolio Impact by Asset



→ Shows the actual scenario execution:

- Geopolitical scenario
- Impact Score = 8
- Portfolio Before = \$10.9M
- Portfolio After = \$10.15M
- Simulated P&L = -\$749.7K
- Loss = -6.88%
- Portfolio impact chart

Screenshot 2: the Stress-Test Results table

→ Shows the detailed asset-level output:

- A01–A10
- Asset type
- Market value
- Shock %
- Stressed value
- Loss

Stress-Test Results

asset_id	asset_type	market_value	shock_pct	stressed_value	loss
A01	Equity	1800000	-0.1143	1594285.7143	-205714.2857
A02	Equity	1200000	-0.1143	1062857.1429	-137142.8571
A03	Bond	1600000	-0.0229	1563428.5714	-36571.4286
A04	Bond	900000	-0.0229	879428.5714	-20571.4286
A05	Loan	1300000	-0.0457	1240571.4286	-59428.5714
A06	Loan	800000	-0.0457	763428.5714	-36571.4286
A07	Derivative	700000	-0.0686	652000	-48000
A08	Derivative	500000	-0.0686	465714.2857	-34285.7143
A09	Equity	1100000	-0.1143	974285.7143	-125714.2857
A10	Loan	1000000	-0.0457	954285.7143	-45714.2857

RiskPulse AI is a hackathon prototype using synthetic/public data and scenario assumptions for demonstration. It is not investment advice.

WHY IT MATTERS

RiskPulse AI demonstrates how unstructured information can be converted into structured risk intelligence for portfolio analysis.

Potential applications:

- Portfolio monitoring
- Early-warning systems
- Scenario analysis
- Credit-risk monitoring
- Market-event triage
- Risk analyst decision support

VALUE

Information → Risk Signal → Scenario → Portfolio Impact

The prototype emphasizes transparency so that each risk signal can be traced back to its detected event and impact drivers.

CURRENT LIMITATIONS

- Hybrid keyword-based event classification
- VADER is a general-purpose sentiment model
- Synthetic social and portfolio data
- Illustrative stress-shock assumptions
- No production-grade streaming infrastructure

NEXT STEPS

- Fine-tuned financial NLP / FinBERT
- Named-entity recognition
- Event deduplication
- Confidence calibration
- Streaming ingestion
- Model monitoring
- Historical backtesting
- More granular portfolio risk factors

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